

Appendix A: Hands-on Labs Reference

This appendix outlines the hands-on exercises and labs available for each chapter. The complete source code, starter files, and solutions are available in the book's companion repository:

[building-with-llms-companion/exercises](#)

Chapter 1: Intro to GenAI & LLMs

- Parameter Intuition
- Tokenizer Explorer

Chapter 2: The LLM Landscape

- Cost Calculator
- Model Selector

Chapter 3: Working with LLM APIs

- CLI Q&A
- Hello LLM

Chapter 4: Capabilities & Limitations

- Bug Triager
- Code Explainer
- Hallucination Audit

Chapter 5: Prompt Fundamentals

- Prompt Rewriter
- System Message Lab

Chapter 6: Prompting Techniques

- Resume Parser
- Sentiment Classifier
- Technique Shootout

Chapter 7: API Parameters

- Parameter Explorer
- Streaming Chat
- Temperature Lab

Chapter 8: Iteration & Evaluation

- Eval Harness
- LLM Judge
- Refactor Assistant

Chapter 9: Conversation Design

- Context Manager
- Multi-Turn Chatbot
- Persona Lab

Chapter 10: Embeddings & Vector Databases

- Chunking Lab
- Embedding Explorer
- Vector DB Comparison

Chapter 11: RAG Architecture

- RAG API
- RAG Evaluator
- RAG Pipeline

Chapter 12: Security & Guardrails

- Moderation Layer

Chapter 13: Cost Optimization

- Caching Lab

Chapter 14: Ethics & Evaluation

- Eval Harness

Capstone 1: Issue Classifier & Work Order Parser

- See the capstone-1 directory for the mid-point project guidelines.

Capstone 2: Intelligent Fleet Assistant

- See the capstone-2 directory for the final project guidelines.